

HAT-P-12b

R-0886

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-12_180618 · created 2026-07-29T01:13:02+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458287.7154 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.162 +/- 0.016
Transit depth (Rp/Rs)^2	2.61 +/- 0.5 [%]
Orbital Inclination (inc)	87.32 +/- 0.62
Airmass coefficient 1 (a1)	1.181 +/- 0.016
Airmass coefficient 2 (a2)	-0.128 +/- 0.019
Scatter in the residuals of the lightcurve fit is	1.35 %
Best Comparison Star	None
Optimal Aperture	3.54
Optimal Annulus	15.12
Transit Duration (day)	0.0885 +/- 0.0063

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.162 ± 0.016 vs 0.1406 (deviation 1.34σ)
Duration fit vs published	0.0885 d vs 0.0975 d (deviation 1.43σ)
Mid-time O–C	-3.7 min
Depth SNR	10.1
Residual scatter	1.35 %

Light curve

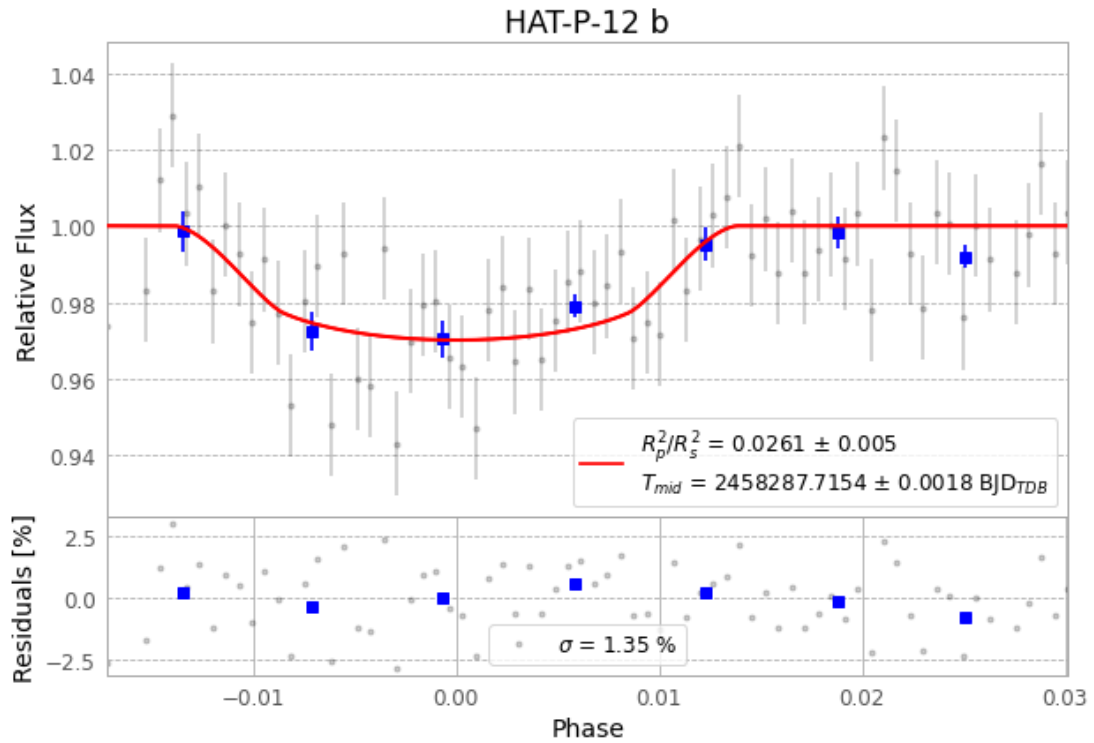


Figure 1 — FinalLightCurve_HAT-P-12 b_2018-06-17.png (SHA-256 daf7b29534889d09...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [373, 270], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8858791198080bf3...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

72 FITS frames (47 MB), HATP-12180618034814.FITS → HATP-12180618072712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-12-180618.log (65ccac18db1e...), FinalLightCurve_HAT-P-12 b_2018-06-17.png (daf7b2953488...), FinalLightCurve_HAT-P-12 b_2018-06-17.csv (2ecc61acf0ed...)

Compute resources

Wall time 110.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 01:13Z.